

Algebra Qualifying Exam Fall 2020

This is just an intro page so you know what is coming. You don't need to start your clocks until you turn the page.

Take this test in an uninterrupted 3 hour chunk, *without* using any outside resources.

Here is the format of what will come. There are six long problems. I'm not expecting people to finish six problems, but I'm expecting people to finish three problems, and get a smattering of other points.

Each problem will be worth 20 points. In addition, each will be graded with a completion score from 0 to 5. This completion score will then be translated into a numerical score according to the table below. Thus the problems are really worth up to 35 points each. When assigning a completion score, we will typically choose 6 main ideas or tasks which need to be overcome in a problem, and give a point for each one. So, if you miss one, you can still get a score of 5.

Finally, the overall test will be graded with a correctness score from 0 to 5, trying to measure what percentage of the stuff you write is actually correct. This correctness score will then be translated into a numerical score according to the same table.

Good luck!

Test specific notes:

Unless otherwise stated, all representations are finite-dimensional with base field $\mathbb{C}$.

There is a character table in problem 6, parts of which I would like you to fill in. If you are not printing out the test, you are welcome to give yourself extra time to copy down this character table.

For an ideal I and a positive integer $n > 0$, I^n is the ideal $I \cdot I \cdot \dots \cdot I$.

Converting completion/correctness scores to numerical scores:

- A 0 is worth 0 points.
- A 1 is worth 3 points.
- A 2 is worth 5 points.
- A 3 is worth 7 points.
- A 4 is worth 11 points.
- A 5 is worth 15 points.

Overview of problems:

- Problem 1: algebras defined by generators and relations.
- Problem 2: linear algebra, modules over PIDs.
- Problem 3: finite groups, semidirect products, conjugacy classes of matrices.
- Problem 4: Jacobson radical, artinian and noetherian.
- Problem 5: Algebraic geometry, nulstellensatz.
- Problem 6: Representation theory, character tables, actions on sets.

1. Let $a \in \mathbb{C}$ be an arbitrary scalar. Let A_a be the (non-commutative) $\mathbb{C}$ -algebra generated by the symbols s, t , modulo the relations

$$s^2 = s + 2, \quad (1a)$$

$$t^2 = t + a, \quad (1b)$$

$$sts = tst. \quad (1c)$$

Note: This problem will be more difficult if you do not know certain techniques for dealing with algebras defined by generators and relations.

a) If $a \notin \{2, 0\}$, prove that $A_a = 0$. Prove that A_2 is six-dimensional, and find a basis.

b) The algebra A_2 acts on $M = \mathbb{C}^2$ by the formulas

$$s \mapsto \begin{pmatrix} 2 & \sqrt{2} \\ 0 & -1 \end{pmatrix}, \quad t \mapsto \begin{pmatrix} -1 & 0 \\ \sqrt{2} & 2 \end{pmatrix}. \quad (2)$$

Prove that this A_2 -module M is irreducible.

c) Prove that any irreducible A_2 -module is at most three-dimensional. (Hint: choose an s -eigenvector.)

2. Let L be a linear endomorphism of a finite-dimensional $\mathbb{F}$ vector space V . Suppose that L has characteristic polynomial

$$p = (x + 1)^4(x - 3)^2(x^2 + 1)^2 = x^{10} - 2x^9 - 7x^8 + 14x^6 + 36x^5 + 62x^4 + 64x^3 + 49x^2 + 30x + 9$$

and minimal polynomial

$$q = (x + 1)(x - 3)^2(x^2 + 1)^2 = x^7 - 5x^6 + 5x^5 - x^4 + 7x^3 + 13x^2 + 3x + 9.$$

a) When $\mathbb{F} = \mathbb{C}$, find the matrix for L in Jordan canonical form.

b) What are the invariant factors of the matrix $xI - L$ (a matrix with polynomial entries)? Find the matrix for L in rational canonical form.

c) Prove that there is a polynomial $f(x)$ such that $f(L)$ projects from V to the (-1) -eigenspace of L . (You need not find it explicitly. Hint: Chinese remainder theorem.)

d) Prove that there is no polynomial $g(x)$ such that $g(L)$ projects from V to the 3-eigenspace of L .

3. This problem will classify all finite groups of order $363 = 3 \cdot 11^2$. Let C_k denote the cyclic group of order k .

a) Prove that any group G of this order is a semi-direct product.

b) How many elements does $\text{Aut}(C_{11} \times C_{11}) = GL_2(\mathbb{F}_{11})$ have? How many elements does $\text{Aut}(C_{11^2})$ have? Justify your answer.

c) How many conjugacy classes are there of matrices $A \in GL_2(\mathbb{F}_{11})$ with $A^3 = I$? Give one matrix from each conjugacy class.

d) List the isomorphism classes of groups of order 363, and justify your answer.

4. On the Jacobson radical.

- a) Give a definition (or any equivalent description) of the Jacobson radical $J(R)$ of a ring R , and use it to prove that any nilpotent element in a commutative algebra is inside its Jacobson radical.
- b) Give an example of a nilpotent element in a non-commutative algebra which is not inside its Jacobson radical.
- c) Give an example of a homomorphism $f: A \rightarrow B$ of commutative $\mathbb{C}$ -algebras such that $f^{-1}(J(B)) \not\subset J(A)$. You need not prove your assertion, but you should identify what $J(A)$ and $J(B)$ are.
- d) Give an example of a homomorphism $f: C \rightarrow D$ of commutative $\mathbb{C}$ -algebras such that $f(J(C)) \not\subset J(D)$. You need not prove your assertion, but you should identify what $J(C)$ and $J(D)$ are.
- e) Suppose R is Noetherian, and let $J = J(R)$. If $J^n = J^{n+1}$ for some n , prove that $J^n = 0$.
- f) Let $f: E \rightarrow F$ be a ring homomorphism, and suppose that E is left Noetherian and Artinian, and F is commutative. Prove that $f(J(E)) \subset J(F)$.

5. Let $R = \mathbb{F}[X]$ be the coordinate ring of an algebraic set, over an algebraically closed field $\mathbb{F}$. If $I, J \subset R$ are two ideals, we define $(I : J) = \{r \in R \mid rJ \subset I\}$. It is easy to prove that $(I : J)$ is an ideal.

- a) Prove that $\mathcal{V}(I : J)$ contains $\mathcal{V}(I) \setminus \mathcal{V}(J)$.
- b) Suppose that I is a radical ideal. Prove that $\mathcal{V}(I : J)$ is the closure of $\mathcal{V}(I) \setminus \mathcal{V}(J)$.
- c) It is fairly easy to prove for ideals I, J, K that $(I : J + K) = (I : J) \cap (I : K)$. When I is a radical ideal, what is the corresponding topological statement about closed sets in the Zariski topology? (You need not prove this topological statement.)
- d) Let $I \subset \mathbb{F}[x, y]$ be the ideal $I = (y - x^2)$. Does there exist an ideal J such that $\mathbb{F}[x, y]/(I : J)$ is finite-dimensional but nonzero (or equivalently, $\mathcal{V}(I : J)$ is a finite non-empty set of points)? If so, find such a J . If not, prove it can't happen.

6. This exercise focuses on the complex representation theory of the finite group $G = GL_2(\mathbb{F}_3)$. Here is its character table: for each conjugacy class we give a representative matrix, and indicate the order of that matrix. Here, I represents the identity matrix.

C	I	$-I$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$
$ C $	1	1	α	8	8	6	β	γ
order	1	2	2	3	6	4	8	8
$\mathbb{1}$	1	1	1	1	1	1	1	1
det	1	1	-1	1	1	1	-1	-1
X	3	3	1	0	0	-1	-1	-1
Y	3	3	-1					
Z_1								
Z_2								
Z_3								
W								

You may freely use the fact that the commutator subgroup $[G, G]$ is equal to $SL_2(\mathbb{F}_3)$, which is the kernel of the determinant map $\det: G \rightarrow \{\pm 1\}$.

- The group G acts on the set $P = \mathbb{P}_{\mathbb{F}_3}^1$ of lines inside $\mathbb{F}_3^2$. Prove that the complement to the trivial representation inside the linearization $\mathbb{C}P$ is irreducible and nontrivial. (Recall: if G acts on a set X , then there is an isomorphism of vector spaces between $\text{End}_G(\mathbb{C}X)$ and $\mathbb{C}[X \times X]^G$.)
- Continuing, compute the character of this complement, and explain why it agrees with X in the table above. You need only justify the computation of any two non-identity entries in the table.
- Explicitly construct another three-dimensional irreducible representation Y (not isomorphic to X), and compute its character.
- Compute the dimensions of the remaining irreducible representations.
- Fill in the column associated to $-I$.

Note: You do not need to fill in the entire character table.